

Sritiz Sahu

Full-Stack AI Engineer

Jalandhar, Punjab

India

+91 9920433172

sritizsahu07@gmail.com

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Professional Summary

High-performance Software Engineer specializing in the intersection of **Full-Stack Development** and **Agentic AI**. Experienced in architecting scalable production systems (MERN, FastAPI) for 5,000+ users and deploying state-of-the-art AI workflows using **LangGraph** and **Computer Vision**. Proven track record in reducing system latency by 80% and automating complex administrative workflows through intelligent software design.

Education

2024–2028 **B.Tech in Instrumentation & Control Engineering, NIT Jalandhar**
GPA: 8.0/10.0 (Minor in Computer Science)

Technical Skills

AI & Agents	LangGraph, Agentic AI, LLM Function Calling, RAG, Computer Vision, OpenCV, SHAP, Keras
Full-Stack	React.js, Node.js, FastAPI, Express.js, Redux Toolkit, Tailwind CSS, ShadCN UI, Framer Motion
Back-end/Cloud	Firebase, Socket.io, MongoDB, MySQL, RESTful APIs, JWT, RBAC, Swagger, Git
Fundamentals	Data Structures & Algorithms (C++), Operating Systems, DBMS Indexing, System Design, OOP

Experience

- Mar 2026 – Present **Full-Stack AI Intern, VaniAgent (Solvea Technologies), Remote (Bangalore)**
- **Autonomous Agents:** Architecting stateful multi-agent workflows using **LangGraph** to automate complex decision-making processes, reducing manual intervention in SaaS operations.
 - **AI Integration:** Developing **LLM function calling** layers to connect generative models with real-time external APIs, enhancing agent tool-use and reasoning capabilities.
 - **Scalable UI:** Building responsive frontend components in React to visualize agent reasoning paths and state transitions, providing users with a transparent view of AI logic.
- Aug 2025 – Present **Full-Stack Developer, NIT Jalandhar TPO Website Team, Jalandhar**
- **Faculty Dashboard:** Engineered a comprehensive, admin-facing Faculty Dashboard from scratch using the **MERN Stack** to automate recruitment logistics for 100+ firms, reducing manual administrative workload by 40%.
 - **Admin Dashboard Enhancements:** Implemented critical UI/UX updates and feature enhancements for the primary Admin Portal, streamlining the student verification and placement tracking workflows.
 - **Performance & Security:** Optimized MongoDB aggregation pipelines to achieve an **80% reduction** in query latency and architected secure middleware using **Bcrypt/JWT** for industry-standard data isolation.
- Aug 2025 – Present **Backend Developer, Xceed (Official NITJ Web Dev Club), Jalandhar**
- **High Traffic:** Scaled registration platforms for **3,000+ students**, maintaining **99.9% uptime** during peak traffic bursts via modular Express.js controllers and optimized state management.
- Dec 2025 – Jan 2026 **Research Intern, IIT Patna, Patna**
- **CV Optimization:** Benchmarked 10+ SOTA architectures (Attention UNet, MDPNet), identifying critical architectural bottlenecks that improved segmentation accuracy by **12%**.

Featured Projects

- Jan 2026 **SmartDocs | AI-Powered Document Digitization, MERN, OCR, MongoDB GridFS**
- **Purpose:** Eliminates manual data entry by automating text extraction from unstructured legacy PDF documents and images.
 - **Tech Implementation:** Built a scalable storage engine using **MongoDB GridFS** to chunk and stream 500+ binary assets, preventing server-side memory overflows.
 - **AI Pipeline:** Implemented an asynchronous OCR pipeline using **Tesseract.js** with custom heuristic data normalization, achieving **94% digitization accuracy**.

- Feb 2026 **Trust Guard | Explainable AI Fraud Detection**, *React, FastAPI, Scikit-learn, SHAP*
- **Purpose:** Bridges the "black-box" gap in AI by providing human-readable justifications for automated financial fraud detection decisions.
 - **Tech Implementation:** Engineered a **FastAPI** backend with a **Scikit-learn** anomaly detection pipeline, achieving a **15% increase** in detection sensitivity.
 - **XAI Visualization:** Developed a **React.js** dashboard integrating **SHAP** to provide visual, feature-level transparency for 100% of automated fraud alerts.
- Feb 2025 **Utkansh | 3D Cultural Platform**, *React, Spline, Framer Motion*
- **Purpose:** Modernized the digital identity of NIT Jalandhar's cultural fest to maximize user engagement and streamline event registrations for 5,000+ users.
 - **Interactive UI:** Leveraged **Spline** for 3D web-graphics and **Framer Motion** for high-fidelity animations, increasing session duration by **30%**.
 - **Optimization:** Achieved a **Lighthouse score of 90+** through advanced asset optimization and lazy-loading techniques.
- Dec 2025 **Lets Chat | Real-Time Communication Platform**, *React, Firebase, Socket.io, Tailwind*
- **Purpose:** Enables instantaneous global communication through a secure, low-latency bidirectional messaging architecture.
 - **Tech Implementation:** Integrated **Socket.io** for sub-second, event-driven communication and **Cloud Firestore** for persistent message history and real-time state synchronization.
 - **Security & Scale:** Utilized **Firebase Authentication** for secure user onboarding and architected a NoSQL schema to handle high-frequency concurrent chat streams.

Achievements

- Hackathons **National Level Selection:** OpenAI Hackathon (State Representative); **SIH:** Shortlisted for internal round.
- Mathematics **RMO 2018 Qualifier:** Qualified for the Regional Mathematical Olympiad after clearing the PRMO.
- Scholastic **Spell Bee:** 2x National Level Qualifier; **Saevus Olympiad:** 3rd Rank in School (Eco Achievers).
- Technical Led TPO Dashboard to production in 14 days; Active contributor to university open-source infrastructure.